

OSW26BZ06-DV025 - Compact Passive Radar

OSD Reliance 21 | SBIR | Release 6 | Opens Sep 23, 2026 (TPOC direct contact ends) | DSIP Q&A closes Oct 7, 12:00 ET | Proposals due Oct 21, 12:00 ET

Recommendation: PURSUE only with an RF partner; otherwise HOLD. Phase I wants a simulation plus a basic lab prototype, which is feasible on COTS software-defined radios, but the X-band SAR/GMTI requirement and the Phase II miniaturized hardware are well beyond a software team alone. Confidence LOW-MEDIUM.

Phase I award	\$314,363, 12 months
Proposal format	Technical volume 20 pages
Compliance	ITAR/EAR clause. CMMC Level 2 (Self).
Phase I deliverable	System architecture, low-cost receiver components, antenna configurations, edge-processing hardware selection; proof-of-concept algorithms; 'a robust simulation and a basic laboratory prototype'
Phase II	Miniaturized hardware prototype on a small UAS or ground node; outdoor passive detection, tracking and imaging; unit-cost and SWaP validation

What they want

[TOPIC TEXT] "a Small Form Factor (SFF) passive X-Band radar featuring both Synthetic Aperture Radar (SAR) imaging and Ground Moving Target Indicator (GMTI) capabilities... light enough for integration onto Group 1 or 2 UAS, loitering munitions, or unattended ground sensors." 100% receive-only; "the sensor shall not transmit any RF energy."

Phase I algorithms named: reference-signal isolation, clutter cancellation, target detection/tracking under constrained compute. The end product is a simulation and a benchtop prototype validating detection from signals of opportunity within SWaP-C limits.

Phase III commercial uses named: low-cost air-traffic monitoring, counter-UAS for critical infrastructure, airspace monitoring where adding active RF is undesirable.

How we would build it

- **Simulation:** bistatic geometry, illuminator waveform models, clutter and direct-path interference; cross-ambiguity processing and CFAR detection; SAR/GMTI feasibility analysis for candidate X-band illuminators.
- **Lab prototype:** a coherent multi-channel COTS SDR (4-5 channel), reference and surveillance antennas, and our processing chain running on a Jetson-class edge board; detect a cooperative moving target using a local broadcast illuminator to prove the chain, then analyze the X-band case.
- **Edge compute budgeting:** profile the pipeline to size the attritable processor.
- **Architecture study:** component costs, antenna form factors for Group 1/2 UAS, and a path to a custom multi-channel X-band receiver in Phase II.

Resources we would use

- Multi-channel coherent SDRs (KrakenSDR-class, USRP) with public passive-radar toolchains and tutorials.
- Principles of Modern Radar Vol. 3 and Adaptive Radar Resource Management (both cited) - the sponsor's reference frame.
- Academic passive bistatic radar literature (FM/DVB-T/cellular illuminators; X-band passive SAR studies).
- RF/antenna partner for the X-band front end and Phase II miniaturization.

Bottlenecks and risks

- **X-band illuminators of opportunity are scarce:** most public passive-radar work uses FM/TV/cellular; X-band requires cooperative or specific illuminators (satellite/marine radars). This could invalidate the SAR/GMTI ask for our approach - check the literature before committing.
- **Hardware in Phase II:** a miniaturized, tightly synchronized multi-channel X-band receiver is real RF engineering.
- **Domain expertise:** bistatic SAR and clutter cancellation are specialist areas.
- **ITAR staffing.**
- **Incumbents:** ESM/SIGINT houses the topic criticizes still have the receivers; our cost angle is the wedge.

Decision factors

- Fit to team: signal-processing software and edge compute - partial; RF hardware - none.
- Dual use: explicitly stated and credible (counter-UAS, airspace monitoring).
- Decisive pre-proposal experiment: with a COTS multi-channel SDR and a public toolchain, detect a moving target off a local broadcast illuminator and measure processing load - proves the software chain, then decide on the X-band question.

Before Oct 7 (Q&A): ask which X-band illuminators of opportunity the sponsor envisions, whether cooperative illumination is acceptable in Phase I/II demos, and whether a lower band is acceptable for the Phase I lab prototype.

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